MILITARY SPECIFICATION

TUBE, STATIC, SELF-ALIGNING: M2

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of static tube entitled Tube, Static, Self-aligning: M2.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-E-5272	- Environmental Testing, Aeronautical and Associated Equipment, General Specification For.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1168	- Lot Numbering of Ammunition.

DRAWINGS

ORDNANCE CORPS

8785815	- Screw, adjusting.
8800030	- Static Tube, Self-aligning: M2.
8800031	- Spider Yoke Assembly.
8800037	- Nose Collar.
8800039	- Bushing.
8800049	- Probe, Static.
8800052	- Body, Nose Cone Assembly.
8800053	- Body, Nose Cone.
8804080	- Container, Metal, Ammunition for Tube, Static.
8805803	- Probe Protector Assembly.
8824890	- Metal Container.
EL-8817787	- Equipment List.

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The static tube shall comply with all requirements specified on the drawing (dwg) 8800030, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. The contractor shall assign serial numbers as required by drawings. The contractor may select any number which complies with the drawing requirement. These numbers shall not be duplicated.

3.5 Leakage. The leakage rate of the static probe assembly shall not exceed 3.00 millimeters (mm) of mercury (Hg) per minute per 10 cubic inch volume when tested as specified in 4.4.1. The rate of leakage is absolute, therefore the rate of pressure change is inversely proportional to the volume.

3.6 Rate of flow. The orifices in the static probe assembly after assembly in the static tube shall return the the system to atmospheric pressure plus or minus (+) 1 mm Hg within 8 seconds or less, when tested as specified in 4.4.2.

3.7 Movement of probe.

3.7.1 Degree of movement. The tip of the probe shall have freedom of movement in all directions of 3.34 inches (6.68 inches diameter) centered about -A- and -C- on drawing. The maximum movement shall be when the mechanical stop on the Vane and Body Assembly contacts the Aft Yoke, at this point the Vane shall not contact the Nose Collar. Tests shall be performed in accordance with 4.4.3.1.

3.7.2 Resistance to movement. The maximum torque to displace the probe from the centerline to the 3.34 inches minimum radius shall not exceed 20 inch-ounces. The maximum torque to return the probe to the centerline shall not exceed 2.0 inch-ounces. Tests shall be performed in accordance with 4.4.3.2.

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3.8 Probe balance. The self-aligning static tube shall be balanced within 0 to 0.798 inch-ounces when tested in accordance with 4.4.4.

3.9 Environmental testing. The static tube shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The static tube shall also comply with 4.3.3.1 before and after each environmental test. Testing shall be as specified in 4.4.5. Tests shall be conducted in the order specified.

3.9.1 Temperature. The static tube shall show no evidence of mechanical damage or deterioration, and shall comply with the requirements of 3.5 through 3.8 after testing as specified in 4.4.5.1.

3.9.2 Humidity. The static tube shall show no evidence of corrosion, water absorption, or potential deterioration because of accumulated moisture, when tested as specified in 4.4.5.2. After humidity testing, the static tube shall comply with 3.5 and 3.7.2.

3.9.3 Vibration. The static tube shall show no evidence of mechanical damage or deterioration, and shall comply with the requirements of 3.5 through 3.8, after testing as specified in 4.4.5.3.

3.9.4 Acceleration. The static tube shall show no evidence of mechanical damage or deterioration, and shall comply with the requirements of 3.5 through 3.8, after testing as specified in 4.4.5.4.

3.9.5 Acceleration shock. The static tube shall show no evidence of mechanical damage or deterioration, and shall comply with the requirements of 3.5 through 3.8, after testing as specified in 4.4.5.5.

3.9.6 Life. The self-aligning static tube shall be mounted in a fixture that displaces the probe 10 ± 1 degrees from the centerline. The probe shall be rotated at this 10 ± 1 degrees about the centerline for 10,000 revolutions at a minimum speed of 25 rpm. After the test, the assembly shall comply with the requirements of 3.5. The test shall be performed in accordance with 4.4.5.6.

3.9.7 Packed container.

3.9.7.1 Leakage. The packed container shall show no evidence of leakage when tested as specified in 4.4.7.1.

3.9.7.2 Vibration. The container shall show no evidence of fracture, chip, damage of internal component, or loosening of the static tube when tested as specified in 4.4.7.2. After testing the static tube shall comply with the requirements of 3.5 through 3.8.

3.9.7.3 Drop. The container shall show no evidence of a fracture, distortion, damages of internal components, or loosening of the static tube when tested as specified in 4.4.7.3. After testing the static tube shall comply with the requirements of 3.5 through 3.8.

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3.10 Workmanship.

3.10.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.10.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.10.3 Burrs. Parts shall be free from burrs.

3.10.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.10.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 3 complete assemblies (see dwg 8800030 and 4.2.1.1 through 4.2.1.3) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.1.1 Static tube. Two of the three samples selected shall be removed from their containers and subjected to all the examinations and tests specified in Table I with the exception of the tests specified in 4.4.7. The two empty containers shall be reserved for the container tests (see 4.2.1.3.).

4.2.1.2 Static tube components. The three sets of components selected shall be inspected for compliance with all drawings and specifications.

4.2.1.3 Container tests. The remainder static tube shall be left in its container and tested as specified in 4.4.7. The two containers reserved from 4.2.1.1 shall also be subjected to these tests using the same static tube.

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4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table 1. Tests shall be conducted in order specified in Table 1.

Table I

First article inspection and tests

Inspection or Test	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)		X
2. Leakage (see 3.5)		X
3. Rate of flow (see 3.6)		X
4. Movement of probe (see 3.7)		X
5. Probe balance (see 3.8)		X
6. Temperature (see 3.9.1)		X
7. Humidity (see 3.9.2)	X	
8. Vibration (see 3.9.3)	X	
9. Acceleration (see 3.9.4)	X	
10. Acceleration shock (see 3.9.5)	X	
11. *Leakage (see 3.9.7.1)		X
12. *Vibration (see 3.9.7.2)	X	
13. *Drop (see 3.9.7.3)	X	

*See 4.2.1.1 through 4.2.1.3.

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

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4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table II and Table III), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table II
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table III
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Static probe (see dwg 8800049) covering a detail of dwg 8800030.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Length overall	Gage
	102. Location of groove	Gage
	103. Width of groove	Gage
	104. Diameter of groove	Gage
	105. Diameter outside	Gage
	106. Diameter of orifice holes (24 holes)	Gage
	107. Thickness of wall, min	Gage
	108. Location of orifice holes	Gage
Minor:		
	201. Finish improper	Visual
	202. Evidence of poor workmanship (see 3.10)	Visual

4.3.2.1.2 Body, nose cone (see dwg 8800053 covering a detail of dwg 8800052).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Inside diameter (0.2812)	Gage
	102. Diameter of vane assembly holes (4 places)	Gage

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103.	Centerline does not intersect longitudinal centerline within 0.005 (4 places)	Gage
104.	Location screw holes (4-40NC-2B)	Gage

Minor:

201.	Finish improper	Visual
202.	Evidence of poor workmanship (see 3.10)	Visual

4.3.2.1.3 Nose collar (see dwg 8800037 covering a detail of dwg 8800030).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Pitch diameter of large threads	Gage
102.	Minor diameter of large threads maximum	Gage
103.	Length of large threads undercut, maximum	Gage
104.	Length to rear of large thread, minimum	Gage
105.	Inside diameter (small)	Gage
106.	Position of set screw	Gage
Minor:		
201.	Evidence of poor workmanship (see 3.10)	Visual
202.	Finish improper	Visual

4.3.2.1.4 Spider yoke assembly (see dwg 8800031 covering a detail of dwg 8800030).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Any component or operation missing	Visual
102.	Adjusting screw not seated properly	Visual
103.	Coating missing from screw	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.10)	Visual

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4.3.2.1.5 Adjusting screw (see dwg 8785815 covering a detail of dwg 8800030).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Pitch diameter of thread, max	Gage
102.	Major diameter of thread, min	Gage
103.	Length to thread, max	Gage
104.	Total length, min	Gage
Minor:		
201.	Finish improper	Visual

4.3.2.1.6 Static tube: self-aligning (see dwg 8800030).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Center of gravity incorrect (see 4.4.6)	Test
102.	Concentricity of nose collar at 0.50 location with nose collar thread and bushing	Gage
103.	Concentricity of nose cone body assembly at 9.34 location with nose collar thread and static probe at 0.64 location	Gage
104.	Concentricity of static probe at 11.95 location with nose collar thread and static probe at 0.64 location	Gage
105.	Concentricity of static probe at 12.58 location with nose collar thread and static probe at 0.64 location	Gage
106.	Concentricity of fins at 5.00 location with nose collar thread and static probe at 0.64 location	Gage
107.	Plane of fin not in plane with center line of static tube	Gage
108.	Planes of fins not mutually perpendicular	Gage

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109.	Inability of forward yoke to move freely in relation to after yoke	Manual/Visual
110.	Hardware loose	Manual/Visual
111.	Burr or foreign matter in orifice holes	Visual
112.	Finish improper	Visual
113.	Any component missing, damaged or loose	Manual/Visual
114.	Braze inadequate	Visual
115.	Marking misleading, missing or illegible	Visual
116.	Evidence of poor workmanship (see 3.10)	Visual
117.	Orifice openings deformed	Visual
118.	Static probe or vanes bent or otherwise deformed	Visual
119.	Evidence of corrosion or fungus growth	Visual
120.	Objective evidence relative to material missing or improper (see 3.1)	Visual
121.	Serialization improper or missing (see 3.4)	Visual

Minor: None defined.

4.3.2.1.7 Bushing (see dwg 8800039 covering a detail of 8800030).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Inside diameter (0.5005)	Gage
102.	Position of headless screw hole	Gage
103.	Finish of inside diameter	Visual/Standard
Minor:		
201.	Evidence of poor workmanship (see 3.10)	Visual
202.	Finish all over improper	Visual/Standard

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4.3.2.1.8 Probe protector assembly, prior to assembling dust cover (see dwg 8805803).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Weld inadequate	Visual
102.	Hardware missing, loose or improperly located	Visual/Manual
103.	Masking missing	Visual
Minor:		
201.	Protective coating missing or inadequate	Visual
202.	Marking missing, misleading or illegible	Visual
203.	Evidence of poor workmanship (see 3.10)	Visual

4.3.2.1.9 Probe protector assembly (see dwg 8805803).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Cover torn, cut or seam loose	Visual
102.	Hardware missing, damaged or loose	Visual/Manual
103.	Marking missing, misleading or unidentifiable	Visual
Minor:		
201.	Color missing or incorrect	Visual
202.	Evidence of poor workmanship (see 3.10)	Visual

4.3.2.1.10 Metal container, prior to sealing (see dwg 8824890).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	End wrench missing, mislocated or improperly attached	Visual

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102. Static tube missing or improperly aligned.....Visual
103. Support center missing.....Visual
104. Allen wrench missing or improperly bolted.....Visual
105. Any packing material missing or mislocated.....Visual
106. Packing list missing or mutilated.....Visual
107. Static tube assembly improperly located or inadequate support in container.....Visual
108. Desiccant bag missing, torn or mislocated.....Visual
109. Gasket missing or mislocated.....Visual

Minor:
201. Evidence of poor workmanship (see 3.10).....Visual

4.3.2.1.11 Metal container (see dwg 8824890).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Cover loose or improperly assembled	Visual/Manual
	102. Car seal missing, unsealed, broken or mislocated	Visual
	103. Container fractured	Visual
	104. Contents loose to the extent of being in jeopardy	Manual
	105. Protective coating of container damaged to the extent of exposing bare metal	Visual
	106. Marking misleading or unidentifiable (incorrect, incomplete, superimposed or otherwise smeared	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.10)	Visual

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

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Test	Test classification
a. Leakage	Major
b. Rate of flow	Major
c. Movement of probe	Major
d. Probe balance	Major
e. Container leakage	Major
f. Vibration	Major
g. Acceleration	Major
h. Temperature	Major
i. Humidity	Major

Of these tests, f, g, h, and i, are considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

4.3.3.1 Leakage, rate of flow, movement of probe, probe balance, and container leakage. Each static tube shall be tested in accordance with 4.4.1 thru 4.4.4 and 4.4.7.1. If any item fails to comply with 3.5 through 3.8 and 3.9.7.1, the item shall be rejected and removed from the lot.

4.3.3.2 Vibration, acceleration, temperature, and humidity. Two sample static tubes shall be selected at random from each lot, and subjected to these tests in the order specified (see 4.4.5.3, 4.4.5.4, 4.4.5.1 and 4.4.5.2). If either static tube fails to comply with the requirements of 3.9.3, 3.9.4, 3.9.1 and 3.9.2, the lot shall be rejected.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL-8817787). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

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4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Leakage. The orifices in the probe shall be temporarily sealed, the static coupling shall be joined to a pressure sensing device, a positive shut off and a source of vacuum capable of obtaining 500 mm of mercury negative pressure. When the temperature of the part and testing equipment has stabilized (see 6.4) at ambient temperature and the pressure within the part and testing equipment has stabilized at 500 mm of Hg negative pressure the positive shutoff shall be closed. The pressure sensing device shall be observed for 10 minutes. During the test the probe shall be rotated once through its maximum limit of travel.

4.4.2 Rate of flow. The orifices in the probe shall be temporarily sealed. The static coupling shall be connected to a pressure sending device, a chamber of 400 + 20 cubic inches volume, a positive stop and a source of 520 + 25 mm of Hg negative pressure. When the temperature of the part and testing equipment has stabilized at ambient temperature and the pressure within the part and chamber has stabilized at 520 + 25 mm Hg negative pressure, the positive shutoff shall be closed. The seal will then be removed from the probe orifices, and the time required for the system to return to atmospheric pressure shall be observed.

4.4.3 Movement of probe.

4.4.3.1 Degree of movement. The self-aligning static tube shall be mounted in a fixture which simulates the mounting characteristics of the missile. The fixture shall establish a zero centerline at the tip of the probe within 0.010 inches of the centerline of the mounting device. A reference line at 3.340 + 0.010 distance from the parallel to the zero centerline shall be established. The probe tip shall pass the 3.340 mark at a minimum of four points in line with the vanes of the assembly. The probe shall then be moved beyond the minimum until the mechanical stop hits. The vanes shall not contact the nose collar. If the probe tip does not reach the minimum point or the vane contacts the nose collar the probe shall be rejected.

4.4.3.2 Resistance to movement. With the self-aligning static tube mounted in the fixture referred to in 4.4.3.1, the probe tip shall be moved to the minimum displacement line. The torque shall not exceed 20 inch-ounces. The probe shall then be moved to the zero point and the torque shall not exceed 2.0 inch-ounces. The displacement from the zero point shall be in four directions approximately midway between the vanes. Any assembly that exceeds the allowable torque shall be rejected.

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4.4.4 Probe balance. The self-aligning static tube shall be mounted in a fixture which simulates the mounting characteristic of the missile. The fixture shall establish a zero centerline at the tip of the probe within 0.010 inches of the centerline of the mounting device. The fixture shall also establish a line that will indicate when the probe is displaced 12.5° upward from the horizontal. This 12.5° shall be computed using mean dimensions of all components. The assembly shall be held vertical and the probe manually aligned to the zero mark on the fixture. The probe shall be displaced to the 12.5° mark and the torque required to so displace it recorded (R1). Place the fixture in a horizontal position and again deflect the probe to the 12.5° mark and record the torque (R2). Subtract R1 from R2 and the result must be within the requirements as stated in 3.6.

4.4.5 Environmental testing (see 3.9).

4.4.5.1 Temperature. The static tube shall be conditioned at -65°F. ± 5°F until the temperature of the tube becomes stabilized (see 6.4). The static tube shall then be removed from the cold chamber and within 1 minute placed in a hot chamber having a temperature of +600° ± 5°F and held therein for 120 ± 20 seconds. Within 1 hour after removal from the hot chamber the static tube shall be tested in accordance with 4.4.1 through 4.4.4.

4.4.5.2 Humidity. The static tube shall be tested in accordance with MIL-E-5272, 4.4.1, procedure I. Test to be performed with H4049 Probe Protector installed. After the humidity test the static tube shall be tested as specified in 4.4.1 and 4.4.3.2.

4.4.5.3 Vibration. The static tube shall be mounted in a fixture simulating actual mounting conditions. The static tube plus fixture shall be vibrated in 10 minute cycles through a frequency range of 5 to 500 to 5 Hertz (Hz). From 5 to 55 (Hz) the test shall be at a double amplitude of 0.02 inches. From 55 to 500 (Hz) the test shall be at a constant acceleration of 5 gravity units (g's). The input vibrations shall be measured on the mounting fixture close to the nose collar. The static tube shall be subjected to one cycle in each of the ten following positions.

Note: Probe is to be loosely supported in the angular positions specified.

1. Vibrations applied in line with longitudinal axis of static tube. Probe free to swivel.
2. Vibration applied as (1). Probe inclined 22° ± 2° from longitudinal axis (any direction).
3. Vibration applied as (1). Probe inclined 22° ± 2° from longitudinal axis and 120° ± 2° from position (2).

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4. Vibration applied as (1). Probe inclined 22° ± 2° from longitudinal axis and 120° ± 2° from positions (2) and (3).
5. Vibration applied at 90° to longitudinal axis of static tube and in line with fin marked "TOP". Probe free to swivel.
6. Vibration applied as (5). Probe inclined 22° ± 2° from longitudinal axis and 90° ± 2° from line of vibration (either direction).
7. Vibration applied as (5). Probe inclined 22° ± 2° from longitudinal axis and 180° ± 2° from position (6).
8. Vibration applied 90° to longitudinal axis of static tube and 90° to fin marked "TOP".
9. Vibration applied as (8). Probe inclined 22° ± 2° from longitudinal axis and in line with line of vibration (either direction).
10. Vibration applied as (8). Probe inclined 22° ± 2° from longitudinal axis and 180° ± 2° from position (9).

After vibration test, the static tube shall be tested in accordance with 4.4.1 through 4.4.4.

4.4.5.4 Acceleration. The static tube shall be mounted in a fixture simulating actual mounting conditions. The static tube plus fixture shall be subjected to a constant acceleration of 40 g's for 10 seconds minimum in both directions of the longitudinal axis, and 25 g's for ten seconds in both directions minimum for each of the other two mutually perpendicular axes. After the acceleration test, the static tube shall be tested in accordance with 4.4.1 through 4.4.4.

4.4.5.5 Acceleration shock. The assembly shall be mounted in a fixture simulating actual mounting conditions. The static tube plus fixture shall be subjected to three 25 g shocks of approximately 10 milliseconds in each of three mutually perpendicular axes. The static tube shall then be subjected to three shocks of 50 g in a forward direction with a pulse duration of 5 milliseconds. The times shall be determined in accordance with 4.4.1 through 4.4.4.

4.4.5.6 Life. The static tube shall be mounted in a fixture which maintains the probe at an altitude of approximately 10 degrees to the mast. The fixture shall rotate the probe so the tip describes a circle. The assembly shall be rotated 10,000 times, at a minimum speed of 25 revolutions per minute (rpm).

4.4.6 Center of gravity (see dwg 8800030). The center of gravity of the movable mass of the static tube, less bellows, shall be determined by any moment method satisfactory to the contracting officer.

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4.4.7 Packed container.

4.4.7.1 Leakage. The leakage test shall be conducted in accordance with instructions on EL-8817787.

4.4.7.2 Vibration. The packed container shall be rigidly mounted to a suitable vibration-producing apparatus. The package shall be vibrated through the range of frequency from 10 to 55 Hz at a constant double amplitude of 0.060 ± 0.006 inches for a duration of 1 hour in each of three mutually perpendicular major axes. Each cycle, 10 to 55 Hz shall have a duration of between 55 to 65 seconds. The frequency shall be varied uniformly throughout the frequency range specified. The static tube shall then be removed from the container and tested as specified in 4.4.1 through 4.4.4.

4.4.7.3 Drop test. The static tube used in the vibration test (see 4.4.7.2) shall be repacked, using the same container and the same packing material. The package shall be permitted to fall freely a total of ten times from a height of 36 inches onto a solid concrete surface of a minimum thickness of 12 inches. The 36 inch drop height shall be the shortest distance from the concrete surface to the nearest point of the container. There shall be no ropes or other suspending media attached to the container during the fall. One chime drop shall be made on each of four simulated diagonally opposite corners for a total of eight drops. One drop shall be made on each of two adjacent side walls (90° travel) for a total of two drops. When the diagonal chime drops are being conducted, the package shall be suspended with the center vertically above the impact point. When the sidewalk drops are being conducted, the sidewalk shall be positioned in a place parallel to the concrete slab. The drop test shall be repeated using new container and new packaging material. The same self-aligning static tube shall be used for the three drop tests. After completion of the test, the static tube shall be removed from the container and tested in accordance with 4.4.1 through 4.4.4.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The static tube shall be packaged in accordance with dwg 8804080 using the specified moisture barrier.

5.1.2 Level C. The static tube shall be packaged in accordance with dwg 8804080 except that the moisture barrier may be omitted.

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5.2 Packing.

5.2.1 Level A and C. The static tube shall be packed in accordance with dwg 8804080.

5.3 Marking. Marking shall be as specified in accordance with dwg 8804080.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

6.5 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-1730 (Rev. 2) dated 24 September 1964.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1135-A191

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Picatinny Arsenal
ATTN: SARPA-QA-N-Q
Dover, New Jersey 07801

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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